Name: Rein Allen M. Capitulo

Role: Quality Assurance (QA) / Tester

Focus: Responsible for verifying API endpoints, database integrity, and cross-platform communication between the Python client and PHP backend.

Phase I: Infrastructure Stabilization and Environment Debugging

In my capacity as the Tester for the Artistryx platform, my primary objective was to architect a stable local testing environment and ensure the seamless integration of the system's multi-language components. The project's initial phase was characterized by significant infrastructure blockers. Upon deploying the Apache and MySQL services via XAMPP, I immediately identified a critical Port 3306 conflict (System Error 10048), which effectively paralyzed the database initialization.

Rather than simply attempting a restart, I utilized advanced terminal diagnostics to perform a process-level investigation. By identifying the specific Process ID (PID) that was erroneously binding to the MySQL port, I was able to clear the "socket hang" and stabilize the server. This intervention was vital, as it prevented a "False Negative" testing scenario where the backend code might have been blamed for what was ultimately an environmental configuration error.

Phase II: Data Layer Resolution and Schema Refactoring

Following the stabilization of the server, my focus shifted to Database Provisioning and Data Layer Resolution. The laboratory faced a major hurdle when the provided artistryx_db.sql schema failed to import due to a versioning mismatch. The source file utilized the utf8mb4_0900_ai_ci collation, which was unrecognized by the MariaDB instance in our local XAMPP setup.

As the lead for this phase, I performed a manual "Sanitization" of the SQL script, refactoring the collation to the more compatible utf8mb4_general_ci standard. I then moved beyond automated imports to manually provision the users and sellers tables. This manual execution allowed me to verify the structural integrity of the database, ensuring that Foreign Key constraints and Unique Indexes (specifically for email and shopName) were strictly enforced. This meticulous setup was necessary to support the complex relational logic required by the backend creators.

Phase III: Functional API Validation and Logic Handshake

The third phase of my testing cycle involved End-to-End Functional Validation using both Python terminal clients and web-based JavaScript interfaces. During initial connectivity tests with client1.py, I encountered a persistent "Server error or cannot connect" message. To diagnose this, I pivoted from application testing to a manual Network Trace via the browser, which revealed a 404 Not Found error. This diagnostic step allowed me to identify that the API endpoint pathing was misaligned with the local directory structure.

By correcting the routing to http://localhost/ADET/src/service.php, I successfully unlocked the Authentication Handshake. I verified the security logic by confirming that the backend correctly utilized password_verify() to compare incoming strings against the bcrypt hashes stored in the users table. Furthermore, I performed Regression Testing on the Seller module, validating that the "Register Shop" (POST) and "Update Shop" (PUT) operations were functioning as intended.

Phase IV: Cross-Platform Interoperability and Final Assessment

The final testing milestone involved Cross-Platform Interoperability. I validated the backend's capabilities as a decoupled Web Service by successfully performing the same registration and login lifecycle through client2.html. This confirmed that our CORS (Cross-Origin Resource Sharing) headers were correctly configured to handle requests from both terminal and browser-based frontends.

In conclusion, my contribution to the Artistryx project went beyond simple bug detection. By systematically resolving infrastructure conflicts, refactoring database schemas, and correcting API routing, I ensured that the project moved from a broken state to a fully integrated, stable e-commerce platform. My work as a Tester provided the necessary verification that the system is not only functional but also secure and ready for professional deployment.